1. Title of the Project

AI Smart Maintenance System for Predictive Failure Detection and Remaining Useful Life (RUL) Estimation

2. Problem Statement

Industrial equipment and machinery are commonly maintained using fixed schedules or reactive approaches, leading to unexpected breakdowns, production downtime, higher maintenance costs, and inconsistent output quality. A machine's output is often the earliest indicator of its health: as equipment drifts out of calibration or is reconfigured for a different product variant, this shows up first as defects in what it produces well before conventional sensor thresholds would trigger an alarm. Piloted on a PCB production line, existing inspection models are trained for a single equipment configuration and degrade sharply once the line is reconfigured for a different product variant, and retraining from scratch each time is costly and slow. There is therefore a need for an intelligent monitoring system that can continuously track equipment output in real time and adapt to a reconfigured setup using only a small amount of new labelled data, through transfer learning and domain adaptation.

3. Why This Particular Topic?

Predictive maintenance is one of the most critical challenges in modern manufacturing. The electronics and semiconductor industry, particularly PCB manufacturing, faces significant losses due to unplanned equipment downtime and quality degradation. Current PCB production lines operate with either fixed-schedule maintenance (which wastes resources) or reactive maintenance (which causes production halts). Equipment failures in PCB manufacturing are costly; a single 8-hour line shutdown can result in lost production of thousands of boards.

We chose this topic because:

1. **Industry Relevance:** PCB production is a high-volume, time-sensitive process where equipment health directly impacts product quality and delivery timelines. Real-time monitoring and early degradation detection can prevent catastrophic failures before they occur.
2. **Technical Complexity:** This project combines multiple advanced techniques, visual anomaly detection and domain adaptation for generalization across equipment types, making it suitable for capstone-level work. It requires deep learning expertise, practical hardware integration, and full-stack system design.
3. **Practical Impact:** The solution is deployable in real manufacturing environments. By implementing visual inspection-based health monitoring, factories can transition from reactive to proactive maintenance without investing in additional sensor infrastructure, making it cost-effective and scalable.
4. **Research Opportunity:** Domain adaptation for cross-equipment generalization is an emerging area. Standard models fail when equipment configurations change, requiring expensive retraining. Our approach reduces adaptation time from weeks to hours, addressing a real industrial bottleneck.
5. **Learning Outcomes:** This project covers end-to-end system development, data collection, deep learning model training, edge deployment on Raspberry Pi/Jetson, backend API design, and interactive dashboards, providing comprehensive exposure to Industry 4.0 practices.
6. **Societal Benefit:** Predictive maintenance reduces resource waste, minimizes unplanned downtime, improves equipment reliability, and enables factories to operate with leaner maintenance teams, directly supporting sustainable and efficient manufacturing.

4. Objectives and Scope of the Project

Primary Objectives:

1. Develop a YOLOv8-based computer vision model trained on a source-domain PCB dataset (2-layer boards) achieving 95%+ defect detection accuracy.
2. Implement an adaptive, defect-aware domain adaptation layer, going beyond standard feature alignment with a selective, class-aware weighting mechanism, to map source-domain features to target-domain features while preserving defect-specific information.
3. Achieve 85–90%+ defect detection accuracy on a target-domain PCB type (multi-layer boards) using only 50–100 labelled images through few-shot fine-tuning, reducing model adaptation time from weeks (standard retraining) to hours.
4. Detects and classifies multiple defect types: solder joint issues, missing components, trace damage, alignment errors.
5. Generate real-time alerts and an interactive dashboard for production teams, visualizing defect trends, classification metrics, and model performance comparison.
6. Estimate equipment degradation and RUL by transforming defect characteristics and trends over production cycles into a temporal equipment-health representation.
7. Keep the adapter lightweight enough to run inference and fine-tuning directly on resource-constrained edge hardware without full model retraining

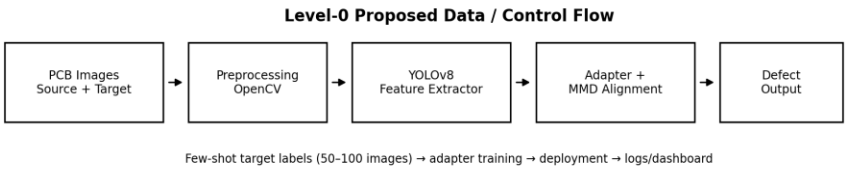
Scope:

1. Use public, annotated PCB datasets (no proprietary data collection required).
2. Implement computer vision-based anomaly detection for manufacturing quality assurance.
3. Demonstrate domain adaptation methodology across at least two PCB types.
4. Develop backend API and frontend dashboard for practical deployment.
5. Document comparative analysis: baseline model vs. domain-adapted model.

5. Methodology

The proposed methodology begins with collecting public PCB defect datasets and organizing them into source and target domains, followed by pre-processing (resizing, normalization, augmentation) using OpenCV. A YOLOv8 detector is trained on the source domain to learn general defect representations, after which a lightweight adapter is introduced between the feature-extraction and detection stages. Feature alignment is enforced using a Maximum Mean Discrepancy (MMD) loss, refined into a weighted, defect-aware variant that aligns defect-relevant regions more strongly, so source and target feature distributions converge while defect-discrimination capability is retained. Only a small labelled target set (approximately 50- 100 images) is required for this adaptation, and the adapted model is evaluated against the source-trained model and a conventional fine-tuned baseline using precision, recall, mAP, and false-positive/false-negative rates. The inference service is exposed through FastAPI, with a camera/edge device supplying the image stream and detection events logged to PostgreSQL; MQTT enables lightweight event communication, and a React/Redux dashboard presents alerts, logs trends, and performance metrics. The adapter and inference pipeline are kept lightweight enough to run directly on resource- constrained edge hardware (Raspberry Pi 4 / Jetson Nano), automatically triggering re-adaptation when the line's configuration changes, without full retraining. As the project currently builds on established techniques (YOLOv8, transfer learning, MMD-based alignment), the research will focus on developing a genuinely novel adaptation mechanism; if this shows measurable technical advancement, a prior-art search and patentability

assessment will follow, with a provisional patent application filed where appropriate before public disclosure.



6. Hardware, Software, and Technology Stack

Programming Languages	Python, JavaScript
Machine Learning / AI	YOLOv8, TensorFlow, PyTorch, Transfer Learning, Domain Adaptation (Feature Alignment, MMD Loss)
Data Processing	OpenCV, NumPy, Pandas, Scikit-learn
Hardware	Industrial camera, Edge device (Raspberry Pi 4 / Jetson Nano), LED lighting setup
Backend	FastAPI, PostgreSQL
Frontend	React, Redux
Communication	REST API, MQTT
Visualization	Matplotlib, Plotly
Datasets	DsPCBSD+, PKU-Market-PCB, PCB-Defect (public PCB defect benchmarks)
Development Tools	Jupyter Notebook, Git, GitHub

7. What Contribution Would the Project Make Towards Society?

1. Reduces unplanned downtime and material waste on manufacturing lines by catching abnormal equipment output earlier and more reliably.
2. Lowers the cost of intelligent equipment monitoring for small and mid-sized manufacturers, who often cannot afford to retrain models every time equipment is reconfigured for a new product variant.
3. Improves the safety and reliability of downstream products (in the pilot domain, electronic devices used in consumer, medical, automotive, and industrial systems) by reducing the chance that a defective unit reaches the field.
4. Provides a reusable equipment-monitoring and domain-adaptation methodology that, beyond the PCB pilot, can extend to other manufacturing equipment and industrial machinery.
5. SDG 9 (Industry, Innovation, Infrastructure): Promotes intelligent manufacturing infrastructure and technological innovation.
6. SDG 12 (Responsible Consumption and Production): Reduces waste through improved quality control and defect detection.

8. The Schedule of the project (Gantt chart)

Activity	W1	W2	W3	W4	W5	W6	W7–W8
Literature review & requirement analysis	✓	✓					
Dataset collection, cleaning & pre-processing		✓	✓				
Source-domain YOLOv8 training			✓	✓			
Adapter + MMD domain adaptation				✓	✓		
Target few-shot evaluation & comparison					✓	✓	
FastAPI + PostgreSQL integration					✓	✓	
React dashboard & alerts						✓	✓
Edge/camera integration & testing						✓	✓
Final evaluation, documentation & paper							✓

9. References/Bibliography

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